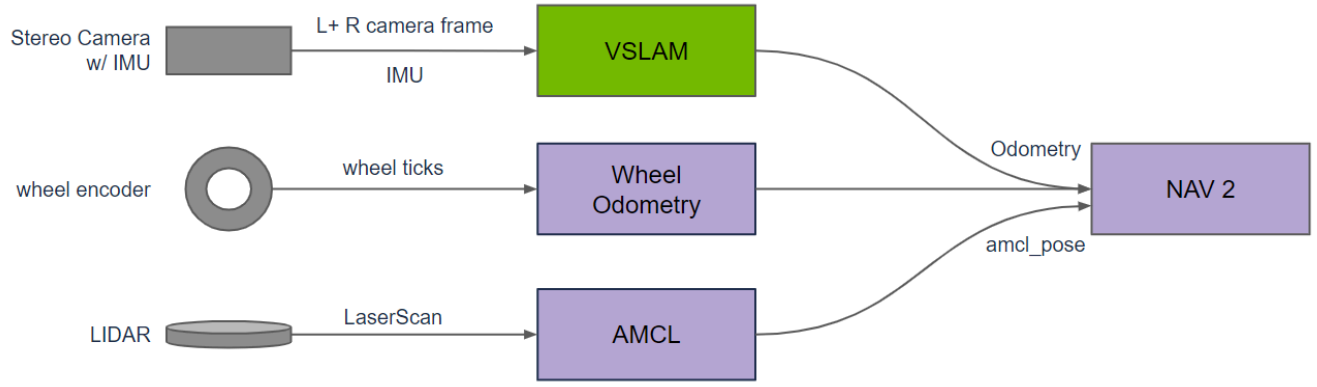
Isaac ROS Visual SLAM

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- Overview
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Isaac ROS Visual SLAM official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_visual_slam/index.html

Overview

Isaac ROS Visual SLAM uses one or more stereo cameras and an optional IMU to estimate the robot odometry, which can be used as the input for the navigation of a mobile robot or the positioning of a drone.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

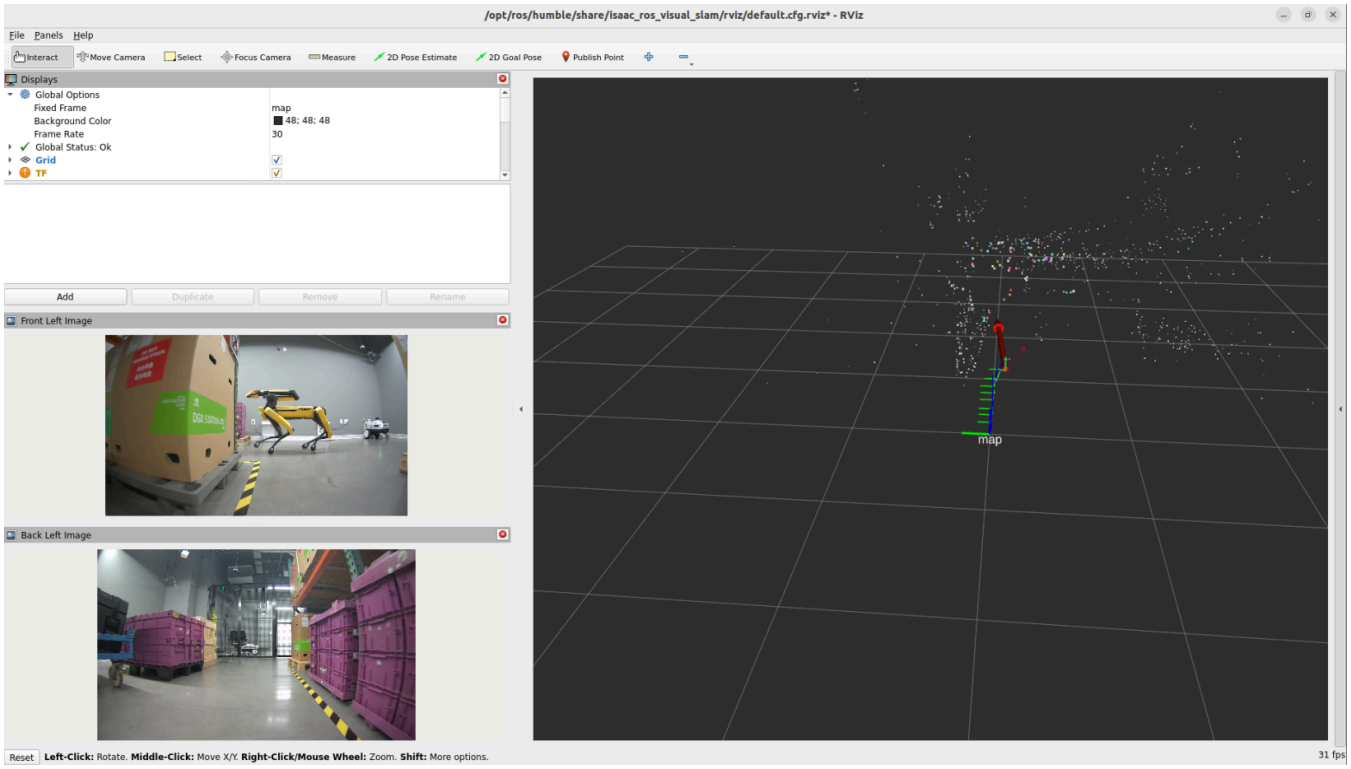
Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
     STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: all the terminals below are executed inside the container:

Terminal A: play the bag in a loop, remapping the front stereo pair to the VSLAM input:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_visual_slam/quickstart_bag \
  --loop --rate 0.5 --remap \
  /front_stereo_camera/left/image_raw:=/visual_slam/image_0 \
  /front_stereo_camera/left/camera_info:=/visual_slam/camera_info_0 \
  /front_stereo_camera/right/image_raw:=/visual_slam/image_1 \
  /front_stereo_camera/right/camera_info:=/visual_slam/camera_info_1
```



Terminal B: listen to the odometry

```
ros2 topic hz /visual_slam/tracking/odometry
```

You can also check a single odometry message and the map output topics:

```
ros2 topic echo /visual_slam/tracking/odometry --once
ros2 topic info /visual_slam/vis/landmarks_cloud -v
ros2 topic info /visual_slam/tracking/slam_path -v
```

Terminal C: start the custom cuVSLAM and the RViz launch:

```
ros2 launch /workspaces/isaac_ros-dev/launch/vslam.launch.py
```

RViz uses `/workspaces/isaac_ros-dev/rviz/vslam_nano.rviz` with `Fixed Frame` set to `map`. It displays the front left image `/visual_slam/image_0`, the rear left image `/back_stereo_camera/left/image_raw`, the SLAM landmarks/path and the single-frame odometry. Do not use `vslam_keepall.cfg.rviz` instead, because it overlays a large number of historical odometry graphics.